Detecting and Parsing Humans

Lecture 19, second part
Notes for Probabilistic Models of the Visual Cortex

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What this lecture covers

Image dependent relations

Parsing occluded people

Summary

Colors used in the figures

These notes follow Yuille's slides "Detecting and Parsing Humans", which these notes call B (Yuille, B). B presents two papers: Chen and Yuille (2014), on articulated pose estimation with image dependent pairwise relations, and Chen and Yuille (2015), on parsing occluded people. A caption citing "B p. *N*" took its picture from page *N* of B.

In the stick figures drawn for these notes and the slides, the color of a limb only tells the limbs apart: the head and neck, the two arms and the two legs each have their own color, and the neck, the root of the tree, is the black node. In the pictures from the papers, the captions say what the colors mark; the colored sticks drawn on photographs are estimated poses, one color per limb.

Pose estimation locates every joint in a single image

The task is to estimate the articulated pose of a person from a single static image. The person is described by parts, which here are joints: elbows, wrists, shoulders, hips, knees, ankles, the head and the neck. The pose is the image location of every joint. Drawing the estimated joints and the limbs between them on the photograph gives a stick figure.

Two tools are combined. DCNNs (deep convolutional neural networks) are very successful at detecting things in images, and can detect the parts. They are, however, black boxes with no explicit representation of the parts and of the spatial relationships between them, so a detector alone may put an elbow far from any shoulder. Graphical models represent the spatial relationships between the parts explicitly: which part is connected to which, and how far apart they are likely to be. The method of this lecture (Chen and Yuille, 2014) trains a DCNN to detect the parts and uses a graphical model to combine the detections.

Pose estimation locates every joint in a single image

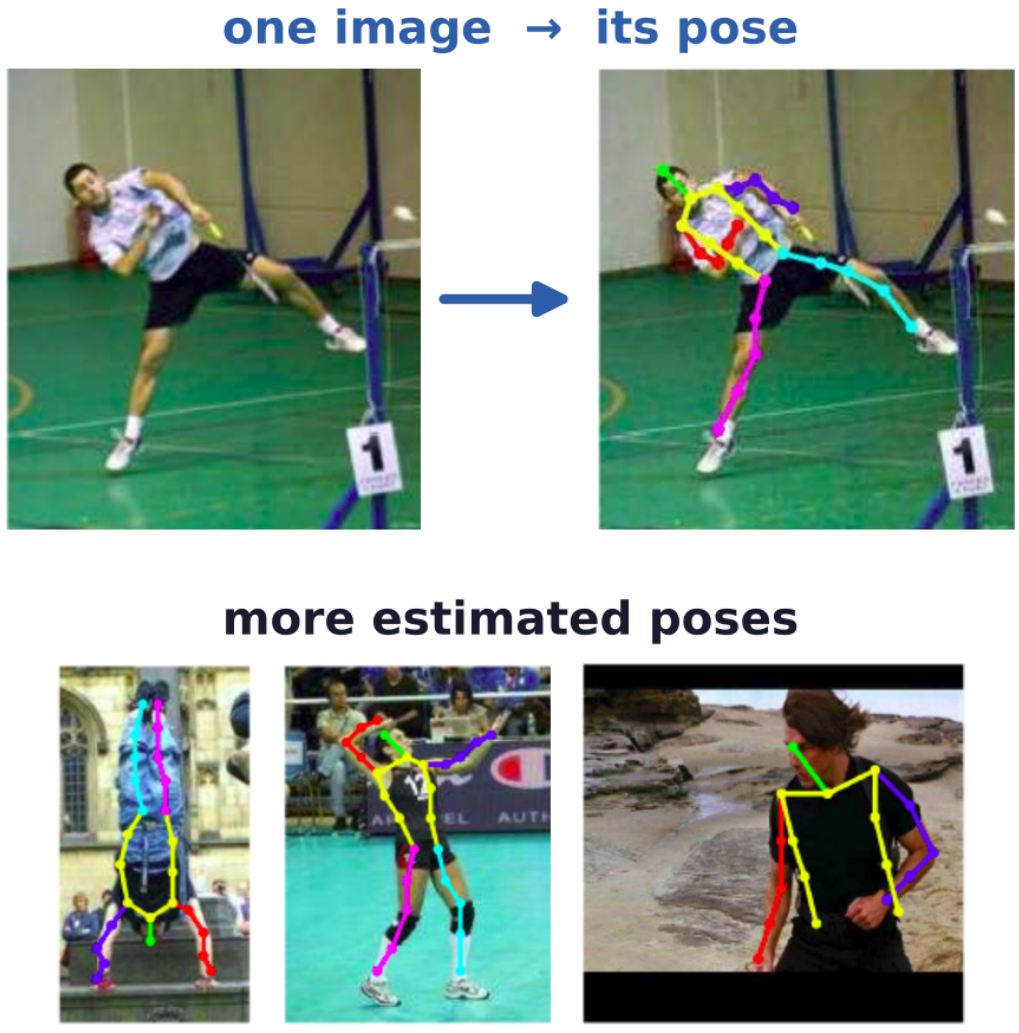


Figure 1: Top: a badminton player and the estimated pose (B p. 5). Bottom: estimated poses of three other people (B p. 7). Each limb is drawn as a colored stick between its two joints.

A local patch predicts where the part's neighbors are

The key observation is that the local image patch around a part reliably predicts the relative positions of its neighbors, as well as the presence of the part itself. A patch around an elbow shows the forearm and the upper arm leaving it in particular directions, so it tells roughly where the wrist and the shoulder are.

The relative position of a neighbor varies, so it is described by a small number of types: the clusters of the relative positions in the training data (an elbow with the wrist below it, an elbow with the wrist to the left of it, and so on). The model therefore has pairwise relations between neighboring parts that depend on the image: the type of each relation is predicted from the local patch. These are the image dependent pairwise relations (IDPRs) of the paper (Chen and Yuille, 2014).

A local patch predicts where the part's neighbors are

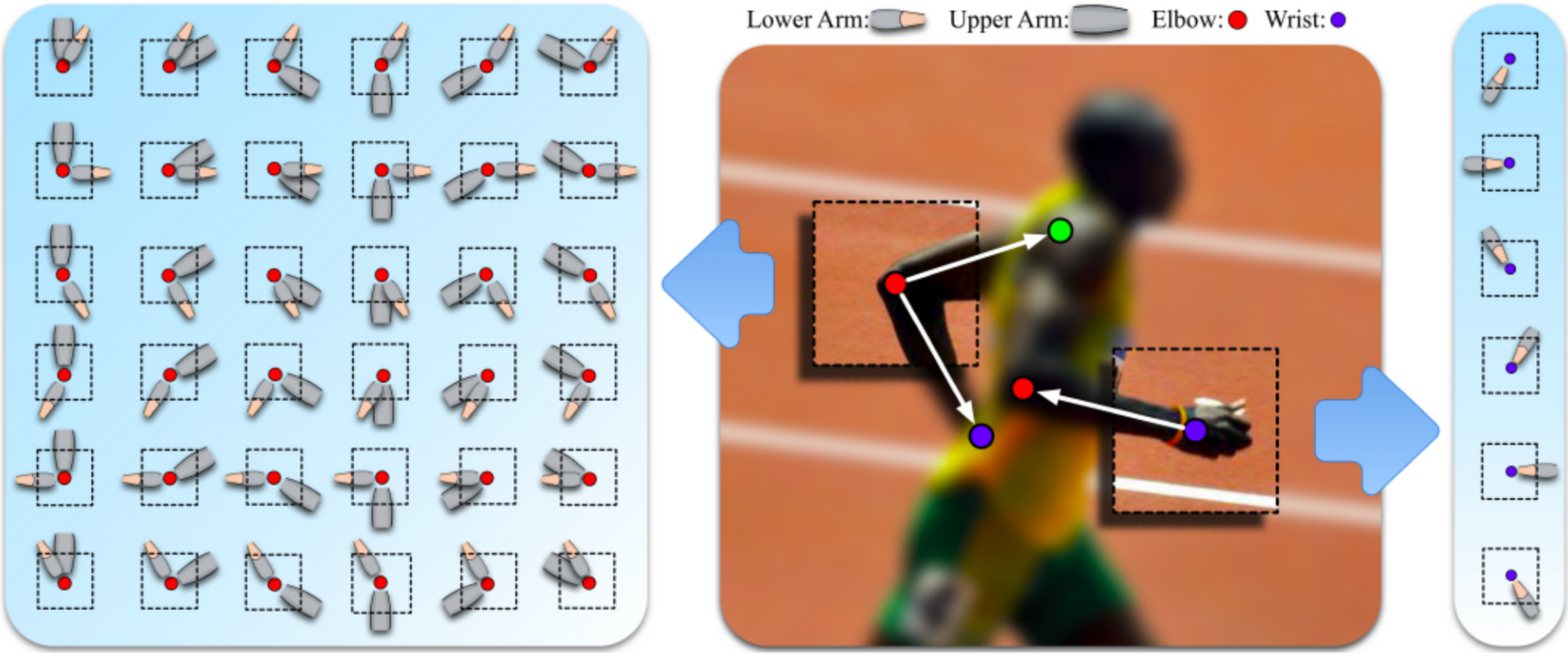


Figure 2: Image dependent pairwise relations (B p. 6; Chen and Yuille, 2014, Figure 1). Center: the patch at the elbow (red dot, dashed box) predicts the shoulder (green) and the wrist (purple), and the patch at the wrist predicts the elbow; the white arrows show these predictions. The large panel (36 boxes) shows typical types of the elbow's relations to its neighbors (each box: an upper arm and a lower arm meeting at the elbow); the narrow panel (6 boxes) shows typical types of the wrist.

A tree model scores joint locations and relation types

The graphical model is a tree $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ whose $K = |\mathcal{V}|$ nodes are the parts and whose edges connect neighboring parts. The variables are the pixel location $\mathbf{l}_i = (x, y)$ of each part i, and, for each edge (i, j), the types $t_{ij} \in \{1, \ldots, T_{ij}\}$ of the relation of j relative to i and t_{ji} of i relative to j. The model has $T_{ij} = 11$ types for every edge. To make the limbs better defined, the paper adds the midpoints between neighboring joints as parts: $K = 26$ on the full bodies of LSP (Leeds Sports Poses) and $K = 18$ on the upper bodies of FLIC (Frames Labeled In Cinema), the two datasets of the results.

The unary term scores the appearance of part i at location $\mathbf{l}_i$:

$$U(\mathbf{l}_i \mid \mathbf{I}) = w_i \, \phi(i \mid \mathbf{I}(\mathbf{l}_i); \theta),$$

where $\mathbf{I}(\mathbf{l}_i)$ is the image patch centered at $\mathbf{l}_i$, ϕ is the log-probability, computed by a DCNN with parameters θ, that the patch shows part i, and w_i is a learned weight.

A tree model scores joint locations and relation types

The pairwise term of an edge (i, j) is the image dependent pairwise relational (IDPR) term:

$$R(\mathbf{l}_i, \mathbf{l}_j, t_{ij}, t_{ji} \mid \mathbf{I}) = \langle \mathbf{w}_{ij}^{t_{ij}}, \psi(\mathbf{l}_j - \mathbf{l}_i - \mathbf{r}_{ij}^{t_{ij}}) \rangle + w_{ij}\, \varphi(t_{ij} \mid \mathbf{I}(\mathbf{l}_i); \theta) + \langle \mathbf{w}_{ji}^{t_{ji}}, \psi(\mathbf{l}_i - \mathbf{l}_j - \mathbf{r}_{ji}^{t_{ji}}) \rangle + w_{ji}\, \varphi(t_{ji} \mid \mathbf{I}(\mathbf{l}_j); \theta).$$

Each line belongs to one end of the edge. Its first term is the spatial term: $\mathbf{r}_{ij}^t$ is the mean relative position of j from i for type t, and $\psi(\Delta\mathbf{l}) = [\Delta x \;\; \Delta x^2 \;\; \Delta y \;\; \Delta y^2]^T$ are the standard quadratic deformation features of the deviation $\Delta\mathbf{l}$ from that mean. With weights $\mathbf{w}_{ij}^t$ that make the squared terms negative, the term is a quadratic penalty for deviating from the mean relative position of the type. Its second term is image dependent: $\varphi(t_{ij} \mid \mathbf{I}(\mathbf{l}_i); \theta)$ is the log-probability, computed by the DCNN from the patch at i, that the relation to j has type t_{ij}.

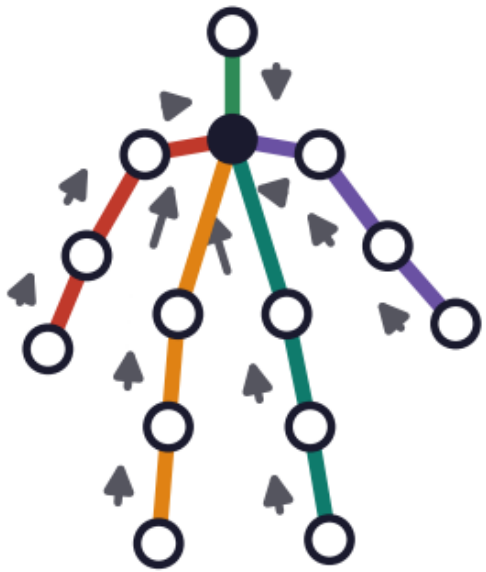
A tree model scores joint locations and relation types

The full score of a pose adds the terms of all the parts and edges and a bias w_0:

$$F(\mathbf{l}, \mathbf{t} \mid \mathbf{I}) = \sum_{i \in \mathcal{V}} U(\mathbf{l}_i \mid \mathbf{I}) + \sum_{(i,j) \in \mathcal{E}} R(\mathbf{l}_i, \mathbf{l}_j, t_{ij}, t_{ji} \mid \mathbf{I}) + w_0.$$

Here $\mathbf{l}$ collects all the part locations and $\mathbf{t}$ all the types. The model's estimate of the pose is the pair $(\mathbf{l}, \mathbf{t})$ with the largest score. The image enters in two ways: through the appearance of each part (ϕ) and through the predicted types of its relations (φ), which tell the spatial terms which mean relative position to use. Both ϕ and φ come from one DCNN.

A tree model scores joint locations and relation types



a tree over the parts;
root: the neck (black)

messages pass from
the leaves to the root

Figure 3: The tree of the model on a stick figure of 14 joints (the model adds the midpoints of the limbs). The neck (black) is the root; the arrows show the direction of the messages of dynamic programming, from the leaves (the head, wrists and ankles) to the root.

One DCNN computes all the image dependent terms

The image dependent terms need information about pairwise part relations, as well as part presence, from local image patches. A DCNN is suitable for this, since it is efficient and shares features between different parts and part relations.

The DCNN classifies a patch $\mathbf{l}(\mathbf{l}_i)$ into one of the elements of a set S. Each element is either a part c together with the types $\mathbf{m}_{c\mathcal{N}(c)}$ of its relations to all its neighbors $\mathcal{N}(c)$, or the background ($c = 0$). The network learns the conditional distribution $p(c, \mathbf{m}_{c\mathcal{N}(c)} \mid \mathbf{l}(\mathbf{l}_i); \theta)$ over S with a softmax output. Its architecture is that of a standard classifier of its time: 5 convolutional layers, 2 max-pooling layers and 3 fully connected layers, applied to patches of 36×36 pixels on LSP and 54×54 on FLIC.

One DCNN computes all the image dependent terms

The two terms of the model are computed from this distribution:

$$\phi(i \mid \mathbf{I}(\mathbf{l}_i); \theta) = \log p(c = i \mid \mathbf{I}(\mathbf{l}_i); \theta),$$

$$\varphi(t_{ij} \mid \mathbf{I}(\mathbf{l}_i); \theta) = \log p(m_{ij} = t_{ij} \mid c = i, \mathbf{I}(\mathbf{l}_i); \theta).$$

The first is a log-marginal: it sums the probabilities over all the type combinations of part i. The second is a log-conditional: m_{ij} is the entry of $\mathbf{m}_{i\mathcal{N}(i)}$ for neighbor j, and φ is the probability, given that the patch shows part i, that its relation to neighbor j has type t_{ij}.

The terms are needed at every location of the image, and they are computed in one pass over the image: the fully connected layers are treated as 1×1 convolutions, so the whole network becomes a convolution over the image, and the computations common to overlapping patches are shared.

One DCNN computes all the image dependent terms

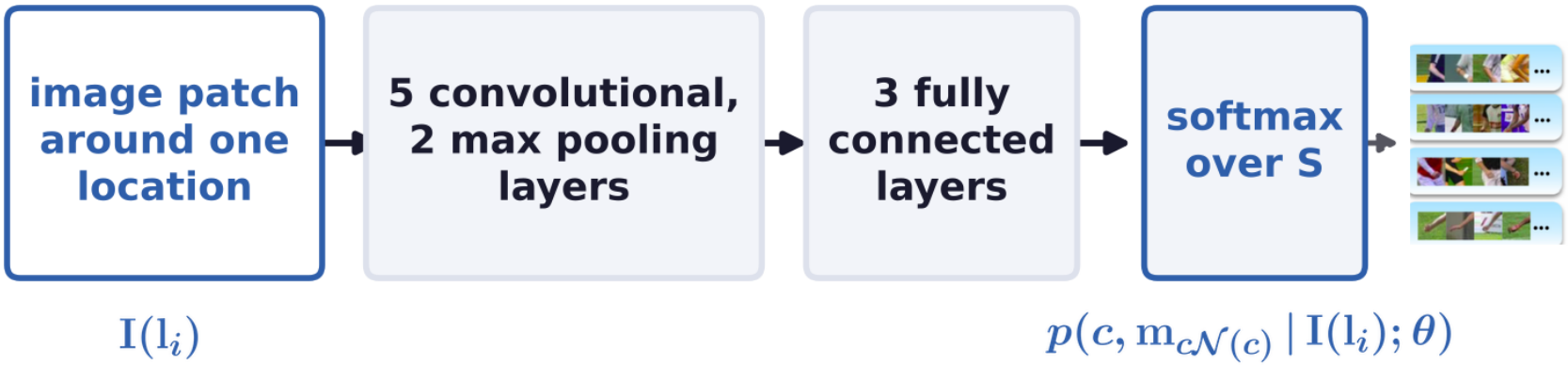


Figure 4: The DCNN of the model (after Chen and Yuille, 2014, section 3): a patch $\mathbf{I}(\mathbf{l}_i)$ goes through convolutional and fully connected layers to a softmax over the set S. The four rows after the softmax are example patches of four elements of S (B p. 10).

Dynamic programming finds the best pose exactly

The estimate is

$$(\mathbf{l}^*, \mathbf{t}^*) = \arg\max_{\mathbf{l},\mathbf{t}} F(\mathbf{l}, \mathbf{t} \mid \mathbf{I}).$$

Because the graph is a tree, the maximum can be found exactly by dynamic programming. Choose the neck as the root. Each part j sends to its parent i the message

$$h_{j\to i}(\mathbf{l}_i) = \max_{\mathbf{l}_j, t_{ij}, t_{ji}} \left[R(\mathbf{l}_i, \mathbf{l}_j, t_{ij}, t_{ji} \mid \mathbf{I}) + U(\mathbf{l}_j \mid \mathbf{I}) + \sum_{k \in \mathrm{ch}(j)} h_{k\to j}(\mathbf{l}_j) \right],$$

where $\mathrm{ch}(j)$ are the children of j: the best score of the subtree below j for each location of i. Messages are computed from the leaves to the root, and each message stores, for every $\mathbf{l}_i$, the maximizing $\mathbf{l}_j$ and types. The root takes the best location given all its messages; starting from it, each child's best location is read from the stored maximizers, down to the leaves.

Dynamic programming finds the best pose exactly

The cost is dominated by the messages. For a fixed pair of types, the bracket of the message is an arbitrary function of $\mathbf{l}_j$ (the unary term, the image dependent term of t_{ji} and the children's messages) plus a quadratic in the displacement $\mathbf{l}_j - \mathbf{l}_i$ with negative squared terms. Its maximum over $\mathbf{l}_j$, for every one of the L locations of $\mathbf{l}_i$, is a generalized distance transform, computed in $O(L)$ time instead of $O(L^2)$ (Felzenszwalb and Huttenlocher, 2005). There are T^2 pairs of types per edge and $K-1$ edges, so the inference takes

$$O(T^2LK)$$

time once the image dependent terms are computed, where L is the number of locations, K the number of parts and T the number of pairwise types.

Dynamic programming finds the best pose exactly

Learning is supervised, from images with annotated joints, in three parts. First, the pairwise type labels are derived from the annotated part locations by clustering: for each edge, the relative positions of j from i in the training data are clustered by k-means into T_{ij} clusters, whose centers are the mean relative positions $\mathbf{r}_{ij}^t$, and every training example gets its type. Second, the DCNN parameters θ are learned with these labels as a classifier over S. Third, the weights $\mathbf{w}$ (all the w's of the score) are learned by a structured SVM (support vector machine): since the score is linear in $\mathbf{w}$ once ϕ and φ are computed, the SVM finds the weights that make the score of the true pose exceed the scores of other poses by a margin.

The model relates to two older ones. With one relation type per edge ($T_{ij} = 1$) it has the structure of a pictorial structure (Fischler and Elschlager, 1973; Felzenszwalb and Huttenlocher, 2005). Restricting each part to predict only the relative position of its parent ($T_{ij} = 1$ if j is not the parent of i) gives the structure of the mixtures of parts of Yang and Ramanan (2011). Those models score the appearance with HOG (histogram of oriented gradients) filters, where this one uses a DCNN.

The model outperforms the state of the art on LSP and FLIC

The model is evaluated on two datasets. LSP (Johnson and Everingham, 2010) has 1000 training and 1000 test images of people doing sports, with full-body poses. FLIC (Sapp and Taskar, 2013) has 3987 training and 1016 test frames from films, with upper-body poses.

The measure is the strict PCP (percentage of correct parts): a part, such as a lower arm, is correct if both its endpoints (the joints at its ends) are within half the part's length of the annotated endpoints; only the single best pose of each image is evaluated. The tables give the PCP of the torso, the head, the upper and lower arms and legs; the mean on LSP is over ten parts, the torso, the head and two of each limb.

The model improves the mean PCP on LSP from 69.2 (Pishchulin et al., 2013) to 75.0, with the largest gains on the lower arms, the hardest parts, and on FLIC from 68.3 to 91.9. DeepPose (Toshev and Szegedy, 2014), which regresses the joint positions with a deep network, used person-centric annotations and 10,000 extra training images (the asterisk).

The model outperforms the state of the art on LSP and FLIC

LSP	Torso	Head	U.arms	L.arms	U.legs	L.legs	Mean
Chen and Yuille	**92.7**	**87.8**	**69.2**	**55.4**	**82.9**	**77.0**	**75.0**
Pishchulin, ICCV	88.7	85.6	61.5	44.9	78.8	73.4	69.2
Ouyang et al.	85.8	83.1	63.3	46.6	76.5	72.2	68.6
DeepPose*	–	–	56	38	77	71	–
Pishchulin, CVPR	87.5	78.1	54.2	33.9	75.7	68.0	62.9
Eichner and Ferrari	86.2	80.1	56.5	37.4	74.3	69.3	64.3
Yang and Ramanan	84.1	77.1	52.5	35.9	69.5	65.6	60.8

FLIC	U.arms	L.arms	Mean
Chen and Yuille	**97.0**	**86.8**	**91.9**
MODEC	84.4	52.1	68.3

Strict PCP (%) on LSP and FLIC (Chen and Yuille, 2014, Tables 1–2; B pp. 14–15). U., L.: upper, lower. Bold: the best in each column. MODEC is the model of Sapp and Taskar (2013); the other rows name the methods as Chen and Yuille (2014) do, whose references give each paper.

The model outperforms the state of the art on LSP and FLIC



Figure 5: Images of LSP, FLIC and Buffy with estimated poses drawn as colored sticks (B p. 16).

Image dependent pairwise terms improve on fixed ones

Two experiments show where the gain comes from and how well the model transfers.

The diagnostic experiment on LSP removes terms. The unary term alone (Unary-Only, the best location of each part from the DCNN, independently) is not powerful enough, even though it is trained by a DCNN classifier: 40.5 mean PCP. The No-IDPRs model keeps the tree with pairwise terms that do not depend on the image; it reaches 64.6, comparable with the state of the art of the time. Adding the IDPR terms raises it to 75.0. Both the tree and the image dependent relations matter; on LSP the image independent pairwise terms add more (24.1 points) than the IDPRs (10.4).

Cross-dataset generalization is tested on the Buffy Stickmen dataset (Ferrari et al., 2008): 276 test frames from a television series. The model trained on FLIC is applied without any training on Buffy (Ours*). The rows without "strict" use the Buffy PCP of the dataset's toolkit, a looser criterion than strict PCP; on the lower arms the model is far ahead under both.

Image dependent pairwise terms improve on fixed ones

LSP	Torso	Head	U.arms	L.arms	U.legs	L.legs	Mean
Unary-Only	56.3	66.4	28.9	15.5	50.8	45.9	40.5
No-IDPRs	87.4	74.8	60.7	43.0	73.2	65.1	64.6
Full model	**92.7**	**87.8**	**69.2**	**55.4**	**82.9**	**77.0**	**75.0**

Buffy	U.arms	L.arms	Mean
Ours*	96.8	**89.0**	**92.9**
Ours*, strict	94.5	84.1	89.3
Yang and Ramanan	**97.8**	68.6	83.2
Yang and Ramanan, strict	94.3	57.5	75.9
Sapp et al.	95.3	63.0	79.2
Karlinsky and Ullman	93.2	60.6	76.9
Eichner et al.	93.2	60.3	76.8

Top: strict PCP (%) on LSP with terms removed (Chen and Yuille, 2014, Table 4). Bottom: PCP (%) on Buffy, Buffy PCP unless marked strict; Ours* is trained on FLIC only; Yang and Ramanan is their TPAMI 2013 version (Table 3). B pp. 17–18. Bold: the best in each column.

In realistic images many object parts are occluded

In real scenes, people are usually significantly occluded by other people or objects, or truncated by the border of the image. In a group photograph most people show only their upper body, and many hide an arm behind a neighbor. A model that requires every part to be present must place the hidden parts somewhere, and it puts them on whatever image structure fits best, often a part of another person. Previous graphical models are robust to only a few occlusions. The method extends the model of Chen and Yuille (2014) to deal with occlusion (Chen and Yuille, 2015).

In realistic images many object parts are occluded

Two observations help. The first is a prior: the observed parts of a person are usually connected. When an arm is hidden, the whole arm beyond some joint is missing, and the visible parts still form a connected piece of the body. The second is that the image itself gives evidence of occlusion. The absence of body-part evidence can help to predict occlusion: the right wrist of the lady in brown can be inferred to be occluded because there is no suitable wrist near her elbow. Absence of evidence is, however, not evidence of absence, and it fails in some scenes: her left arm is completely occluded, yet a nearby person (the lady in green) provides strong image evidence of an elbow and a wrist at plausible locations. The image near an occlusion boundary, around her right elbow and her left shoulder where the arms disappear, also gives evidence of the occlusion.

In realistic images many object parts are occluded

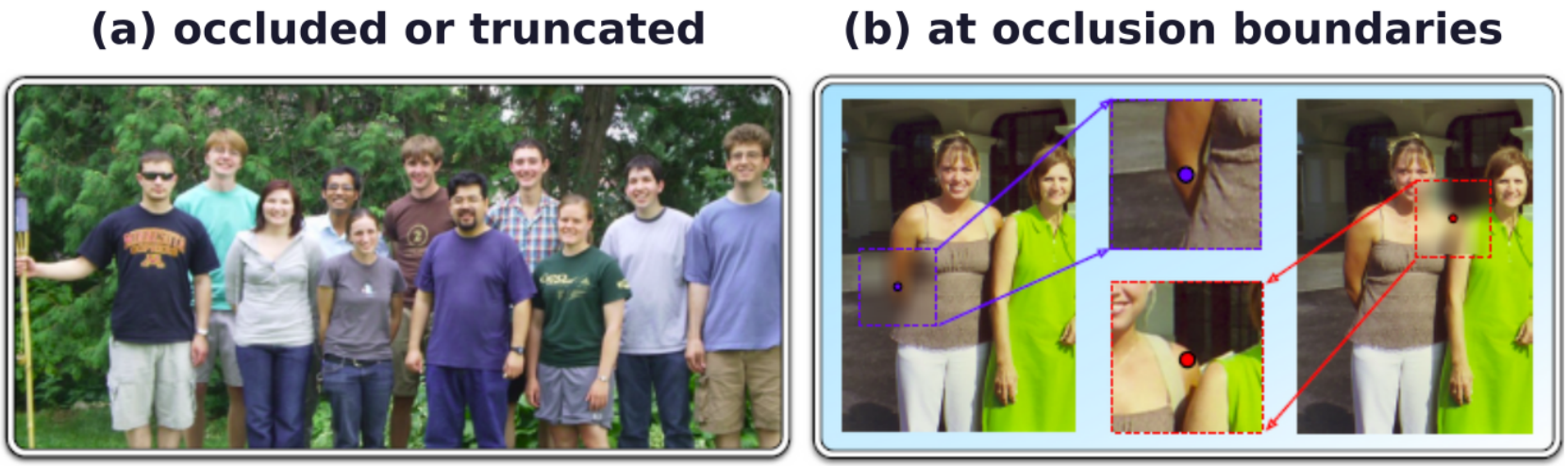


Figure 6: (a) A group photograph in which most people are occluded or truncated. (b) Evidence at the occlusion boundaries of the lady in brown: her right elbow (purple box) and her left shoulder (red box), enlarged in the middle. B p. 21; Chen and Yuille (2015), Figure 2.

Every connected subtree is a flexible composition

The model of Chen and Yuille (2014) is a tree over all the parts: the full graph. A flexible composition is any connected subtree of the full graph, including the full graph itself. A composition that does not contain some parts is suitable for a person with those parts occluded: a person with a hidden left arm is described by the composition without the left shoulder's subtree, and a person cut off at the waist by the composition without the legs.

A composition $\mathcal{G}_c = (\mathcal{V}_c, \mathcal{E}_c)$ has the parts $\mathcal{V}_c$ and the edges $\mathcal{E}_c$ of a connected subtree. The number of compositions can be very large: $N(N+1)/2$ already for a chain of N parts, and far more for a branching tree, so the model cannot score each composition separately. They share their parts, however, and this makes the search efficient.

Every connected subtree is a flexible composition

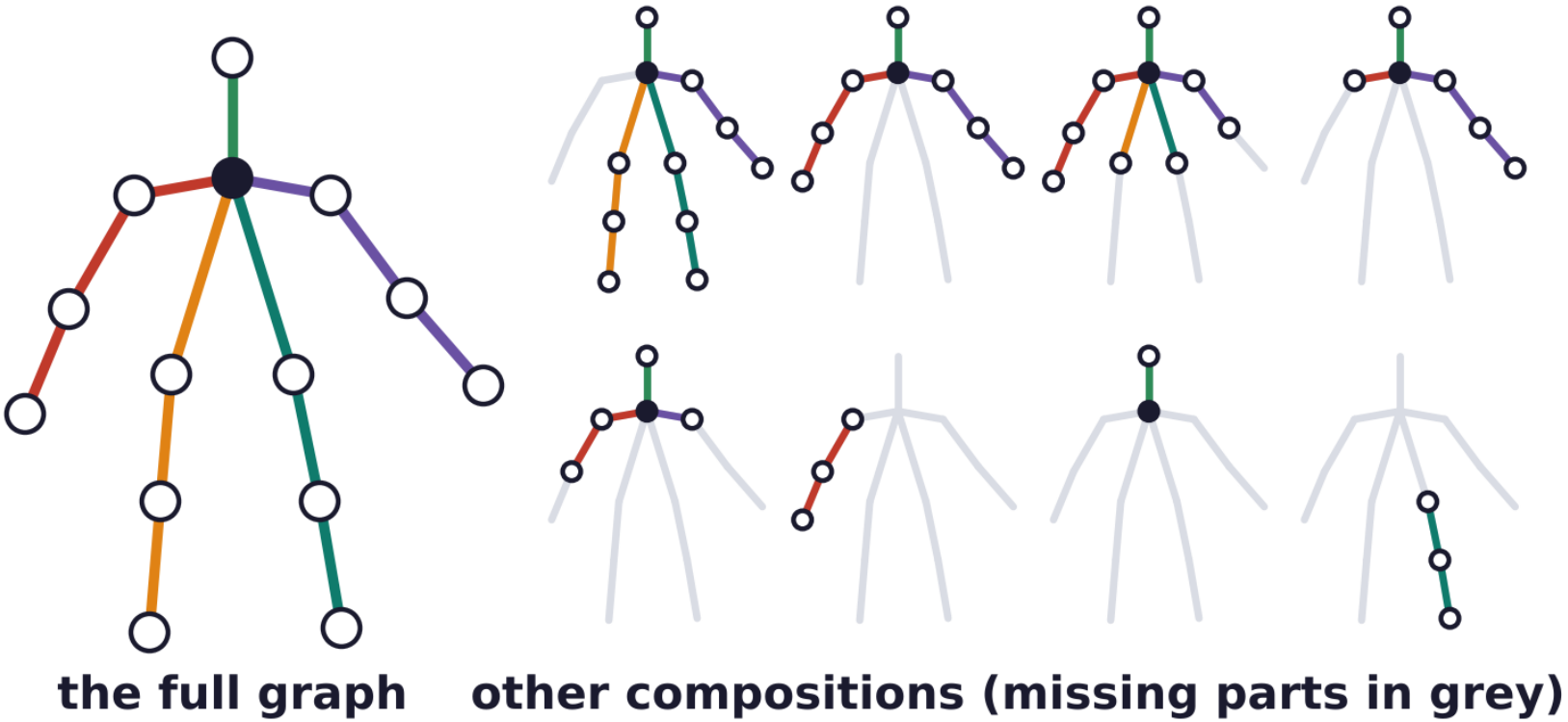


Figure 7: The full graph and eight of its flexible compositions. In each composition the parts it contains are drawn in color and the missing parts in grey; every composition is a connected subtree (after Chen and Yuille, 2015, Figure 1, redrawn).

The local patch also gives evidence of occluded neighbors

The DCNN of the 2014 model predicts, from the patch around a part, the relative positions of its neighbors. Chen and Yuille (2015) add one more outcome for each neighbor: occluded. When the neighbors are visible, the local image measurement around a part, such as the elbow, reliably predicts their relative positions, as in the model of 2014. When a neighbor is occluded, the local image measurement also provides evidence for the occlusion: the arm ends at an occluder, a different texture or the border of another person.

The figure shows the types of the elbow's relations to the wrist and the shoulder, now including types in which the lower arm or the upper arm is hidden by an occluder (the hatched boxes). The DCNN is trained to recognize all of them from the patch at the elbow.

The local patch also gives evidence of occluded neighbors

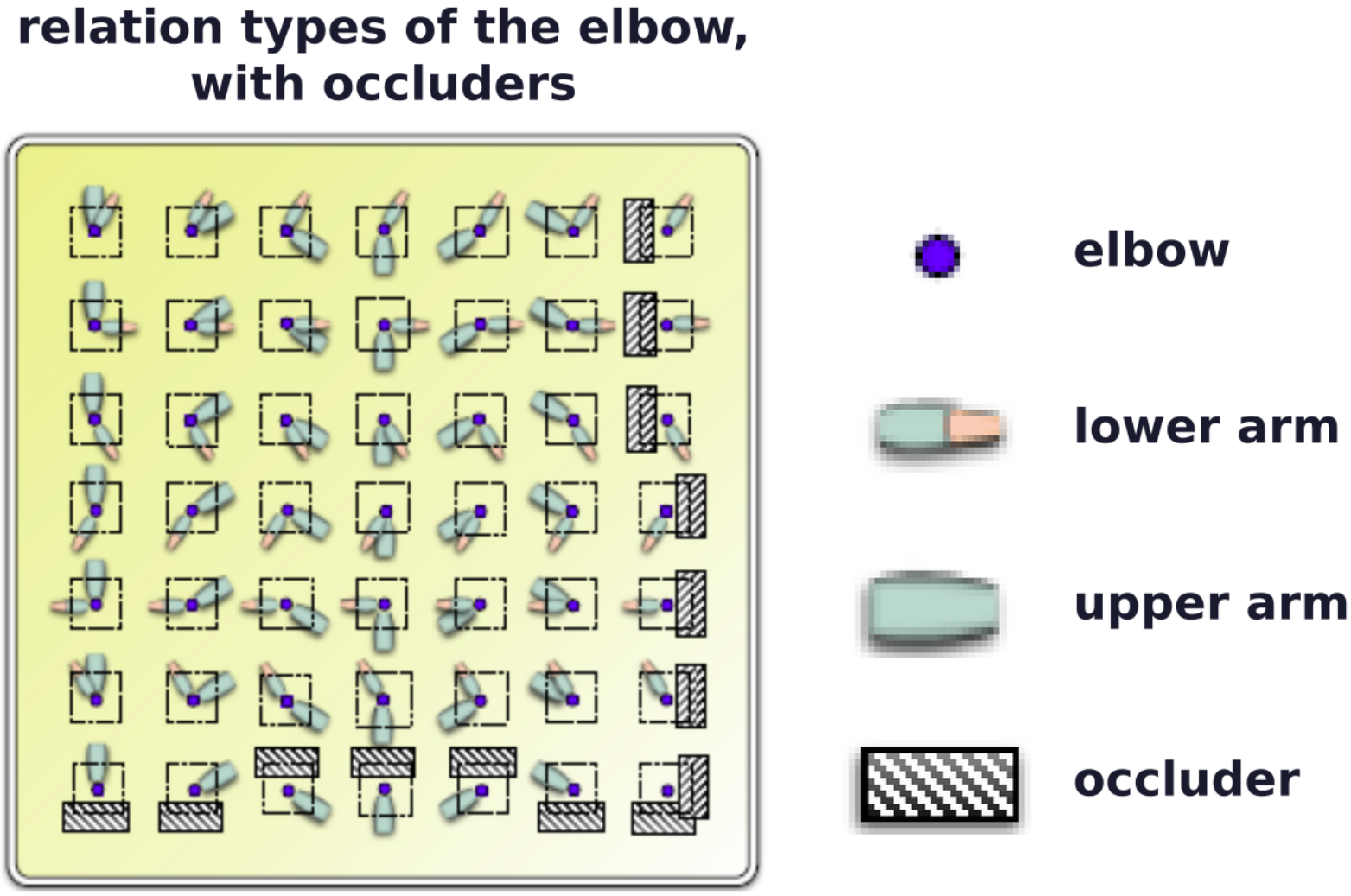


Figure 8: Relation types of the elbow (purple dot): each box shows the lower and upper arm leaving the elbow; in the last column and row one is hidden by an occluder (hatched), in the corner both. B p. 22; Chen and Yuille (2015), Figure 3; legend icons enlarged.

Decoupling terms and missing-part penalties score a pose

A composition is made by cutting edges of the full graph. For a composition $\mathcal{G}_c$, the decoupled edges are $\mathcal{E}_c^d = \{(i,j) \in \mathcal{E} \mid i \in \mathcal{V}_c,\ j \notin \mathcal{V}_c\}$: the edges that join a part i of the composition to a part j outside it. Cutting the edge (i,j) removes the subtree $\mathcal{T}_j$ rooted at j (on the side of the edge away from i), whose parts $\mathcal{V}(\mathcal{T}_j)$ are all occluded. A binary variable $\gamma_{ij} \in \{0,1\}$ marks whether the edge is cut.

Decoupling terms and missing-part penalties score a pose

Two terms score a cut. The decoupling term is the image evidence, at i, that j is occluded:

$$D_{ij}(\gamma_{ij} = 1, \mathbf{I}_i \,|\, \mathbf{I}) = w_{ij}\, \varphi^d(\gamma_{ij} = 1 \,|\, \mathbf{I}(\mathbf{I}_i); \theta),$$

where φ^d is the log-probability, from the DCNN, of the outcome "neighbor j occluded" for the patch at i. These are the image dependent occlusion decoupling (IDOD) terms. The missing-part penalty

$$B_{ij} = \sum_{k \in \mathcal{V}(\mathcal{T}_j)} b_k$$

adds a bias b_k for each part k of the removed subtree: the price of declaring it absent, which balances the absence of good part appearance.

Decoupling terms and missing-part penalties score a pose

The score of a pose $\mathbf{l}, \mathbf{t}$ with a composition $\mathcal{G}_c$ of the full graph $\mathcal{G}$ is

$$F(\mathbf{l}, \mathbf{t}, \mathcal{G}_c \mid \mathbf{I}, \mathcal{G}) = \sum_{i \in \mathcal{V}_c} A(\mathbf{l}_i \mid \mathbf{I}) + \sum_{(i,j) \in \mathcal{E}_c} R(\mathbf{l}_i, \mathbf{l}_j, t_{ij}, t_{ji} \mid \mathbf{I}) + \sum_{(i,j) \in \mathcal{E}_c^d} \big(B_{ij} + D_{ij}(\gamma_{ij} = 1, \mathbf{l}_i \mid \mathbf{I})\big).$$

The first line is the base model, the model of Chen and Yuille (2014), restricted to the composition; its unary term is written A in the paper, the U of the 2014 model. The second line scores the cut edges.

Decoupling terms and missing-part penalties score a pose

Inference maximizes the score over the compositions as well as over the locations and types. In the messages of dynamic programming, each child subtree can either be attached, with its message, or cut, with the score $B_{ij} + D_{ij}$. Compositions that do not contain the root are handled by computing the messages along every edge in both directions, so that any part can be the root of a composition. Because all the compositions share their parts, the best composition and pose are found in twice the time of the base model, $O(2T^2LK)$. Learning is as before: k-means for $\mathbf{r}$, the DCNN for θ, and, in this paper, a linear SVM for the weights and biases.

Flexible compositions improve occlusion prediction and PCP

The method is evaluated on the WAF ("We Are Family") dataset (Eichner and Ferrari, 2010): 525 group photographs with six people per image on average, 350 for training and 175 for testing, with upper-body poses. The model has 15 parts on WAF. Two measures are used. AOP, the accuracy of occlusion prediction, measures how well the visibility of the parts (visible or occluded) is predicted. PCP is the occlusion-aware PCP of the dataset's toolkit, a different criterion from the strict PCP of LSP: a part is correct if its endpoints are on average within half the part's length of the annotation, and an occluded part counts only if it is predicted occluded. The mean PCP (mPCP) printed in the table equals (torso + head + 2 upper arms + 2 lower arms)/6.

Flexible compositions improve occlusion prediction and PCP

The flexible compositions (FC) raise the PCP of the base model by 11.6 points and the AOP by 8.1. Adding the IDOD terms (FC + IDOD, the full model) improves the PCP further to 80.7 and the AOP to 84.9, higher than the multi-person and one-person models of Eichner and Ferrari (2010) and the other methods of the table. The largest gain is on the lower arms.

Flexible compositions improve occlusion prediction and PCP

WAF	AOP	Torso	Head	U.arms	L.arms	mPCP
Chen and Yuille	**84.9**	**88.5**	**98.5**	**77.2**	**71.3**	**80.7**
Multi-Person	80.0	86.1	97.6	68.2	48.1	69.4
Ghiasi et al.	74.0	–	–	–	–	63.6
One-Person	73.9	83.2	97.6	56.7	28.6	58.6
Base model	73.9	81.4	92.6	63.6	47.6	66.1
FC	82.0	87.0	98.6	72.7	67.5	77.7
FC + IDOD	84.9	88.5	98.5	77.2	71.3	80.7

Top: comparison on WAF; bottom: diagnostic experiments, FC + IDOD being the full model (Chen and Yuille, 2015, Tables 1–2; B pp. 25–26). AOP and occlusion-aware PCP, %. Bold: the best of the comparison. The diagnostics' best head score is FC's 98.6.

Flexible compositions improve occlusion prediction and PCP



Figure 9: Results on WAF: the parts that the model infers as visible, and thus estimates, are drawn as colored sticks (B p. 27; Chen and Yuille, 2015, Figure 5).

Detected joints are used to estimate 3D pose and actions

The detection of pose, with and without occlusion, is important for many applications. Two of them were developed with collaborators at Peking University.

Estimating the 3D structure of a person from a single image: Wang et al. (2014) start from the detected 2D joints (step 1). The 3D pose is represented as a sparse combination of basis poses learned from 3D motion-capture data, with limb-length constraints. Given a 3D pose, the camera is estimated so that the projected 3D joints match the 2D joints (step 2); given the camera, the 3D pose is re-estimated from the 2D joints (step 3). Steps 2 and 3 alternate. Because 2D detectors are inaccurate, the reprojection error is measured with an L1 norm, which makes the fit robust; the 3D estimate still depends on the detected 2D joints.

The parsing model has a limitation: objects are represented in terms of joints only, which becomes problematic in some human configurations.

Action recognition: a pose-based representation of actions uses the joints detected in each frame of a video (Wang et al., 2013).

Detected joints are used to estimate 3D pose and actions

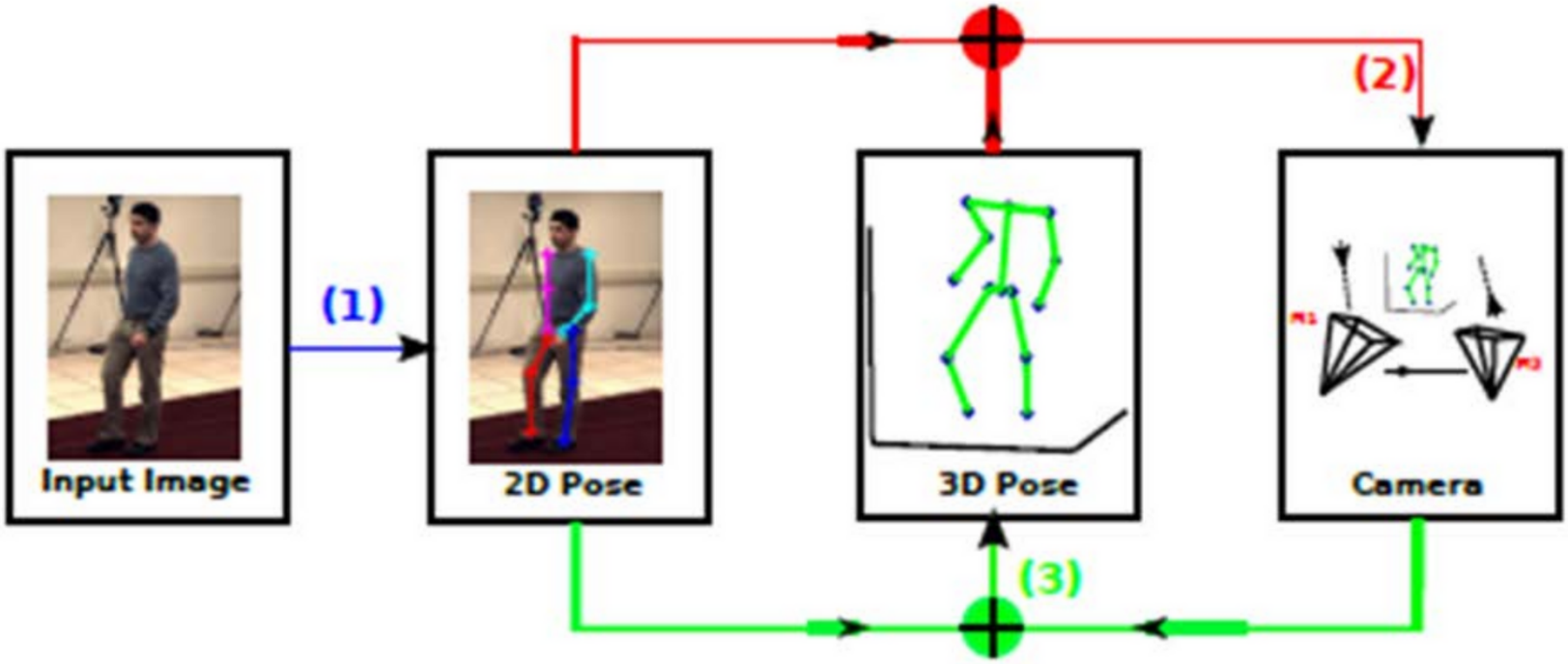


Figure 10: From 2D to 3D (B p. 28; Wang et al., 2014, Figure 1): (1) the 2D pose is detected in the input image; (2) the camera is estimated from the 2D and 3D poses; (3) the 3D pose is estimated from the 2D pose and the camera; the red path (step 2) and the green path (step 3) alternate.

Summary

- Parsing humans: detect the joints, with DCNNs for the parts and graphical models for their relations.
- A tree of parts with pairwise relation types; a local patch predicts where the neighbors are.
- One DCNN gives the appearance and IDPR terms; dynamic programming gives the best pose, $O(T^2LK)$.
- On LSP the image dependent pairwise terms improve on fixed ones; the model transfers to Buffy.
- Flexible compositions handle occlusion: connected subtrees, decoupling terms, missing-part penalties.
- Part sharing keeps inference at twice the cost; the detected joints lead to 3D pose and actions.
- Limitation: objects are represented by joints only, which is problematic for some human configurations.

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